

Announcements :

Calculus A(1)
12/19

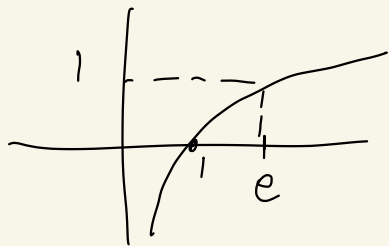
- HW 8 is online (last HW)
- Exercise sheet for training will be posted on Friday.
(we will correct it on 12/26)
- Possibility of taking the final exam this spring

Last time :

$$\log : (0, +\infty) \rightarrow \mathbb{R}$$

$$\log'(x) = \frac{1}{x}$$

$$\log(1) = 0$$



$$\log(e) = 1$$

$(e > 1)$

$$\log(xy) = \log(x) + \log(y)$$

$$\log(x^\alpha) = \alpha \log(x)$$

$$x > 0 \quad \alpha \in \mathbb{Q}$$

$$\exp: \mathbb{R} \rightarrow (0, +\infty)$$

inverse of $\log$

$$\exp = \log^{-1}$$

$$x^\alpha = \exp(\alpha \log(x))$$

if $\alpha \notin \mathbb{Q}$, $x > 0$

we define

$$x^\alpha = \exp(\alpha \log(x))$$

in particular,
we get

$$\boxed{e^\alpha = \exp(\alpha)} \quad \text{if } x = e$$

Basic properties of $\exp$:

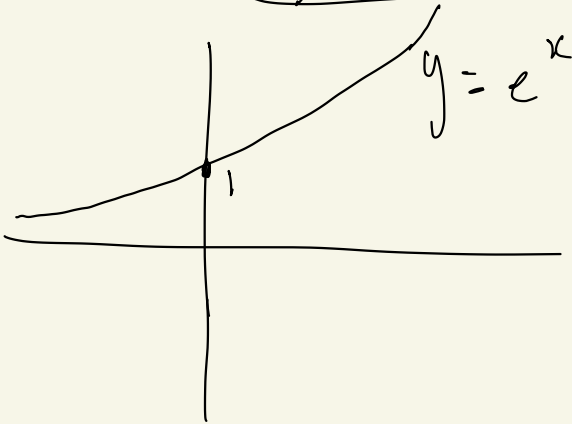
$$\boxed{e^{\alpha+\beta} = e^{\alpha} \cdot e^{\beta}} \quad (\forall \alpha, \beta \in \mathbb{R})$$

$$\boxed{\exp' = \exp}$$

$$\frac{d}{dx} e^x = e^x$$

$\Rightarrow$

$$\boxed{\int e^u du = e^u + C}$$



The power rule (general form)

We saw if $\alpha \in \mathbb{Q}$,

$$\frac{d}{dx} x^{\alpha} = \alpha x^{\alpha-1}$$

Theorem (Power rule)

$$\forall \alpha \in \mathbb{R} \quad / \quad \frac{d}{dx} x^{\alpha} = \alpha x^{\alpha-1}$$

Proof: By definition,

$$x^{\alpha} \stackrel{\text{def}}{=} e^{\alpha \log(x)}$$

$$\text{Chain rule: } \frac{d}{dx} x^{\alpha} = \frac{\alpha}{x} \cdot e^{\alpha \log(x)}$$

$$= \frac{x}{e^{\log(x)}} \cdot e^{x \log(x)}$$

$$= x e^{\log(x)(x-1)}$$

$$\stackrel{\text{def}}{=} x x^{x-1}$$

✱

Ex: $\frac{d}{dx} (1 + \cos(x))^{\sqrt{2}} = ?$

|| Chain rule + power rule. $\sqrt{2}-1$

$$= -\sin(x) \cdot \sqrt{2} (1 + \cos(x))^{\sqrt{2}-1}$$

a^x and $\log_a(x)$

Let $\boxed{a > 1}$. Consider $x \mapsto a^x \stackrel{\text{def}}{=} e^{x \log(a)}$

$$\mathbb{R} \rightarrow (0, +\infty)$$

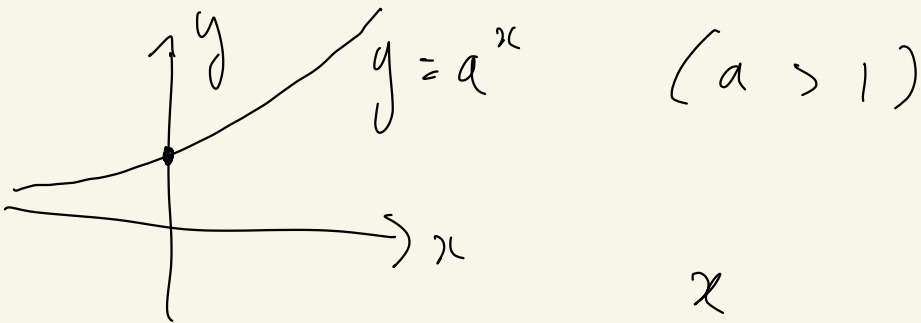
$$\begin{aligned} \frac{d}{dx} a^x &= \frac{d}{dx} e^{x \log(a)} = \log(a) e^{x \log(a)} \\ &= \log(a) a^x \end{aligned}$$

So

$$\int a^u du = \frac{1}{\log(a)} a^u + C$$

Rem: Since $a > 1$, $\frac{d}{dx} a^x > 0$

So $x \mapsto a^x$ is increasing



Fact: $\lim_{x \rightarrow +\infty} a^x = +\infty$

Proof: take $x = n \frac{\log(2)}{\log(a)}$
 $n \in \mathbb{N}$

$$\begin{aligned} a^x &= e^{x \log(a)} = e^{n \log(2)} \\ &= 2^n \xrightarrow{n \rightarrow +\infty} +\infty \end{aligned}$$

Since $x \mapsto a^x$ is increasing,

$$\lim_{x \rightarrow +\infty} a^x = +\infty. \quad \square$$

Fact: $\lim_{x \rightarrow -\infty} a^x = 0$

Proof: $a^x = \frac{1}{a^{-x}}$

So $\lim_{x \rightarrow -\infty} a^x = \frac{1}{+\infty} = 0. \quad \square$

$$x \mapsto a^x \quad \text{one-to-one}$$

$$\mathbb{R} \longrightarrow (0, +\infty)$$

$\log_a =$ inverse of $x \mapsto a^x$

$$\log_a : (0, +\infty) \longrightarrow \mathbb{R}$$

$\uparrow$
called log in base a -

$$\boxed{\log_a(a^x) = x} \quad (\forall x \in \mathbb{R})$$

For $a = e$, $\log_e = \log$

because $e^x = \exp(x)$

$$y = a^x \iff \log(y) = \log(a^x)$$

$$x = \log_a(y) \iff \log(y) = x \log(a)$$

$$\Leftrightarrow x = \frac{\log(y)}{\log(a)}$$

We have: $\forall y > 0$,

$$\boxed{\log_a(y) = \frac{\log(y)}{\log(a)}}$$

We have defined two classes of functions:

$$(1) \cdot x \mapsto a^x \quad \begin{matrix} (a > 1) \\ x \in \mathbb{R} \end{matrix}$$

$$(2) \cdot y \mapsto \log_a(y) \quad \begin{matrix} (a > 1) \\ y \in \mathbb{R} \end{matrix}$$

(1): exponential functions

(2) Logarithm functions

Exponentials grow "very fast"

Logarithms grow "very slow"

Ex: $e^{100} \approx 2.6 \cdot 10^{43}$

$$\log(10^{50}) \approx 115$$

We can make these statements more precise.

Theorem :

① Exponentials grow faster than any polynomial

i.e. $\forall$ polynomial $P(x) = \sum_{k=0}^n a_k x^k$

$$\forall a > 1 \quad n \in \mathbb{N} \quad a_k \in \mathbb{R}$$

We have $\lim_{x \rightarrow +\infty} \frac{P(x)}{a^x} = 0$

② Logarithms grow slower than any power of x

ie. $\forall a > 1$ we have
 $\forall \alpha > 0$

$$\lim_{x \rightarrow +\infty} \frac{\log_a(x)}{x^\alpha} = 0$$

Proof of ① : We apply
L'Hôpital's rule iteratively
if $n = \deg(P)$ ($n \geq 1$)
then $\deg P' = n - 1$

So when we differentiate P
 n times, we get a constant
 On the other hand, if we
 differentiate $x \mapsto a^x$ n -times

we get $\log(a)^n \cdot a^x$

By L'Hôpital,

$$\lim_{x \rightarrow +\infty} \frac{P(x)}{e^x} = \frac{0}{+\infty} = 0$$

Ex : $\lim_{x \rightarrow +\infty} \frac{x^5 + 1}{e^x} = 0$ $\square$

by the theorem.

$$\lim_{x \rightarrow \infty} \frac{\log(x)}{\sqrt{x}} = 0$$

Inverse trigonometric functions

We shall only consider:

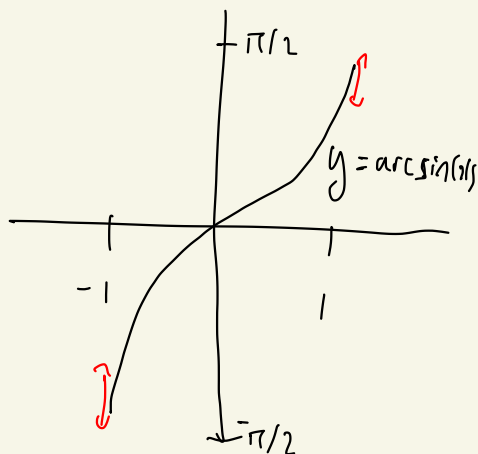
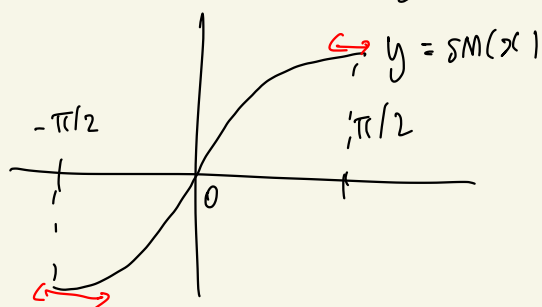
$$\sin, \cos, \tan, \sec = \frac{1}{\cos}$$

and their inverses

$\sin^{-1}$	$\cos^{-1}$	$\tan^{-1}$	$\sec^{-1}$
$\parallel$	$\parallel$	$\parallel$	$\parallel$
$\arcsin$	$\arccos$	$\arctan$	arcsec

Pb: These are not one-to-one
so we restrict their domain.

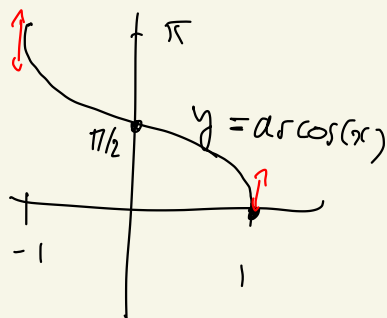
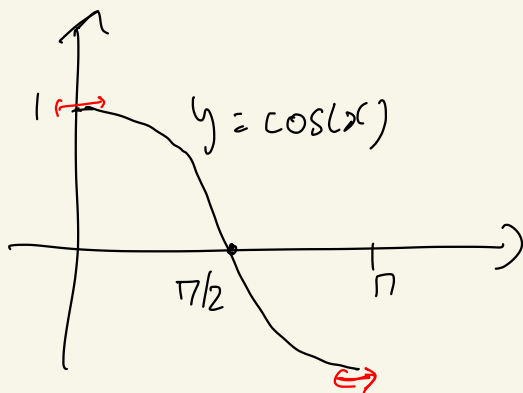
• $\sin : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow [-1, 1]$ one-to-one



$\arcsin : [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$\arcsin$: odd and increasing

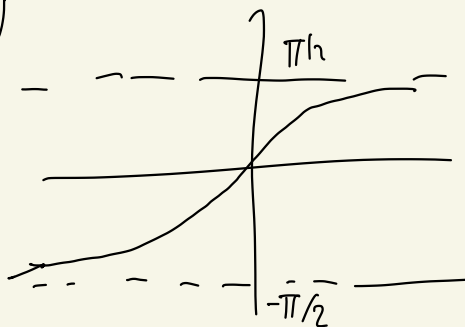
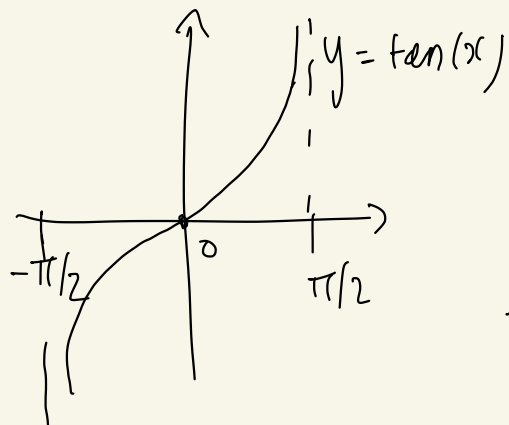
• $\cos : [0, \pi] \rightarrow [-1, 1]$



$\arccos$ is decreasing

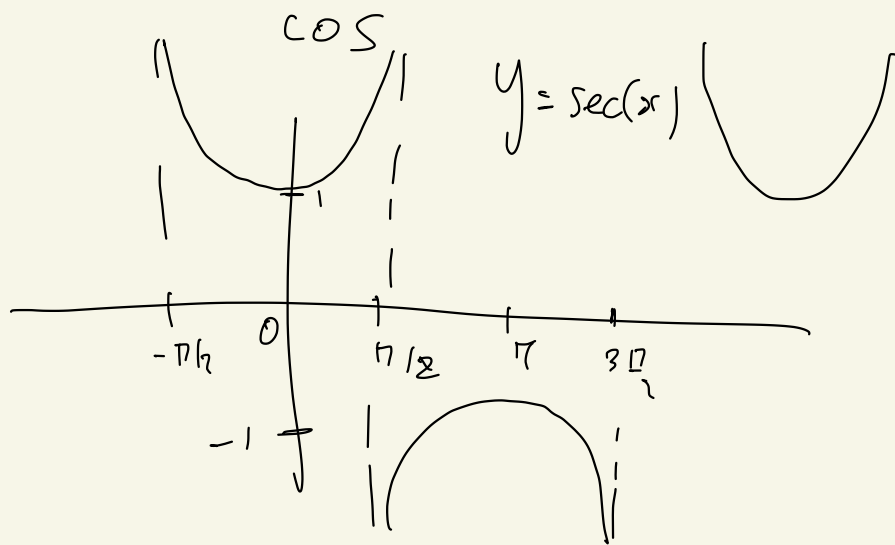
$$\bullet \tan : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$$

$$\arctan : \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$



$\arctan$ is odd and increasing

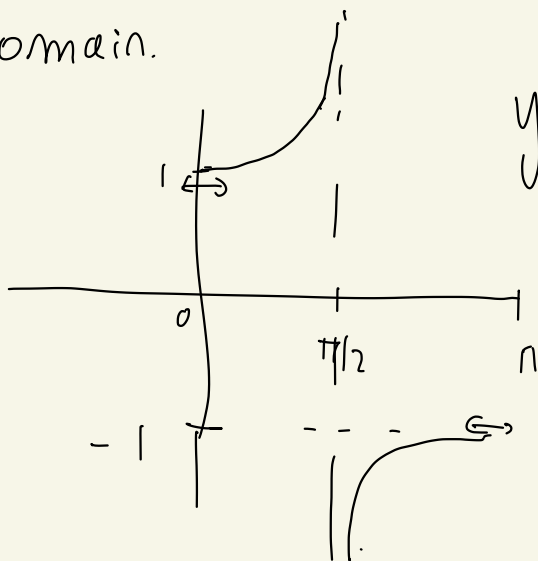
$$\bullet \sec = \frac{1}{\cos}$$



We restrict $\sec$ to

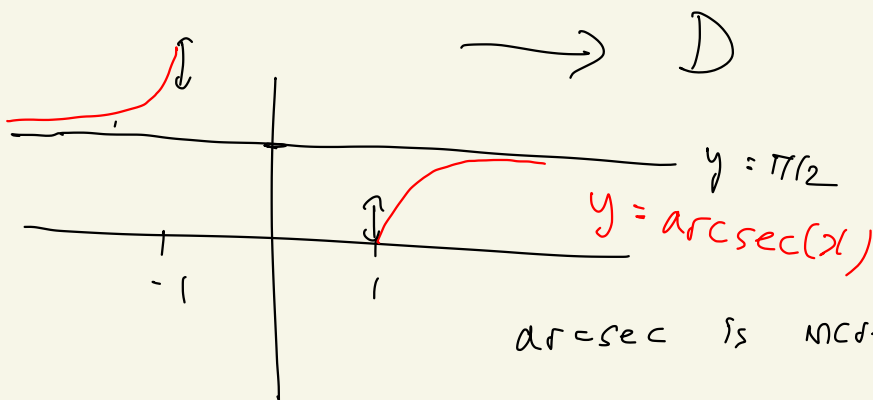
$$D = [0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$$

$\sec$ is one-to-one on that domain.



$y = \sec(x)$
increasing
on D

$$\arcs \sec : (-\infty, -1] \cup [1, +\infty)$$



$\arcs \sec$ is increasing

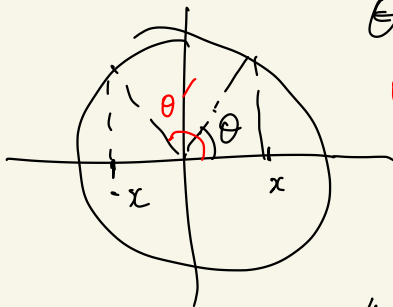
Some special values

x	$-\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{2}}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$
$\arcsin(x)$	$-\pi/3$	$-\pi/4$	$-\pi/6$	$\pi/6$	$\pi/4$	$\pi/3$
$\arccos(x)$	$5\pi/6$	$3\pi/4$	$2\pi/3$	$\pi/3$	$\pi/4$	$\pi/6$

Remark : $\forall x \in [-1, 1]$, we have

$$\arccos(-x) = \pi - \arccos(x)$$

geometric proof : Assume $x \geq 0$



$$\theta = \arccos(x) \in [0, \pi]$$

$$\theta' = \arccos(-x)$$

$$\theta' = \pi - \theta$$

Direct "algebraic" proof :

Let $x \in [-1, 1]$.

$$\arccos(-x) \in [0, \pi]$$

is the unique $\theta' \in (0, \pi]$

$$\text{s.t. } \cos(\theta') = -x$$

$$\text{Let } \theta = \arccos(x) \in [0, \pi] \\ \cos(\theta) = x$$

$$\text{Claim: } \cos(\underbrace{\pi - \theta}_{\in [0, \pi]}) = -x$$

$$\text{So } \theta' = \pi - \theta \text{ by uniqueness}$$

$$\text{Proof claim: } \cos(\pi - \theta)$$

$$= -\cos(\theta)$$

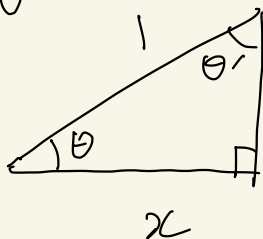
$$= -x \quad \square$$

$$\text{Fact: } \forall x \in [-1, 1],$$

we have

$$\arcsin(x) + \arccos(x) = \frac{\pi}{2}$$

geometric proof (assume $x > 0$)



$$\theta = \arccos(x) \in [0, \frac{\pi}{2}]$$

$$\theta' = \arcsin(x) \in [0, \frac{\pi}{2}]$$

$$\theta + \theta' = \frac{\pi}{2}.$$

Direct "algebraic" proof ($\forall x \in [-1, 1]$)

$$\text{Let } \theta' = \arcsin(x) \in [-\frac{\pi}{2}, \frac{\pi}{2}]$$

$$\text{unique s.t. } \sin(\theta') = x$$

$$\text{Let } \theta = \arccos(x) \in [0, \pi]$$

$$\underline{\text{claim:}} \quad \sin\left(\frac{\pi}{2} - \theta\right) = x$$

If the claim is true, then

$$\text{since } \frac{\pi}{2} - \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}],$$

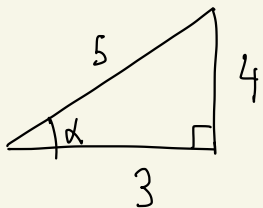
$$\text{we must have } \theta' = \frac{\pi}{2} - \theta.$$

Proof of the claim:

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos(\theta) = x. \quad \square$$

Ex: Assume $\frac{\pi}{2} > \alpha = \arcsin\left(\frac{4}{5}\right) > 0$

Find $\cos(\alpha)$ and $\tan(\alpha)$



$$\cos(\alpha) = \frac{3}{5}$$

$$\tan(\alpha) = \frac{4}{3}$$

Table for special values of arctan

x	$-\sqrt{3}$	-1	$-1/\sqrt{3}$	$1/\sqrt{3}$	1	$\sqrt{3}$
$\arctan(x)$	$-\pi/3$	$-\pi/4$	$-\pi/6$	$\pi/6$	$\pi/4$	$\pi/3$

Derivatives of inverse
trigonometric functions

Very useful to compute many
integrals.

- $\arcsin : [-1, 1] \rightarrow [-\frac{\pi}{2}, \frac{\pi}{2}]$

Note : $\sin : (-\frac{\pi}{2}, \frac{\pi}{2}) \rightarrow (-1, 1)$
 is differentiable and $\sin'$ never vanishes

So $\arcsin$ is differentiable on $(-1, 1)$

$$\frac{d}{dx} \arcsin(x) = \frac{1}{\sin'(\arcsin(x))}$$

$$= \frac{1}{\cos(\underbrace{\arcsin(x)}_{\in (-\frac{\pi}{2}, \frac{\pi}{2})})}$$

$$\cos(\theta)^2 = 1 - \sin(\theta)^2$$

So if $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$

$$\cos(\theta) = \sqrt{1 - \sin(\theta)^2}$$

Since $\arcsin(x) \in (-\frac{\pi}{2}, \frac{\pi}{2})$

we get $\cos(\arcsin(x)) > 0$

And $\cos(\arcsin(x)) = \sqrt{1 - \sin(\arcsin(x))^2}$
 $= \sqrt{1 - x^2}$

So $\frac{d}{dx} \arcsin(x) = \frac{1}{\sqrt{1-x^2}}$

$(x \in (-1, 1))$

So $\int \frac{dx}{\sqrt{1-x^2}} = \arcsin(x) + C$

So $\int \frac{du}{\sqrt{a^2 - u^2}} = \arcsin\left(\frac{u}{a}\right) + C$

$\forall a > 0$

$\forall u: I \rightarrow \mathbb{R}$
 s.t. $|u| < a$
 $\Leftrightarrow u^2 < a^2$

$$(\Rightarrow) a^2 - u^2 > 0$$

Similarly, we get

$$\frac{d}{dx} \arccos(x) = \frac{-1}{\sqrt{1-x^2}}$$

$$\text{i.e. } \int \frac{du}{\sqrt{a^2 - u^2}} = -\arccos\left(\frac{u}{a}\right) + C$$

(This follows also from the relation $\arcsin(x) + \arccos(x) = \pi/2$.)

• We have already seen that

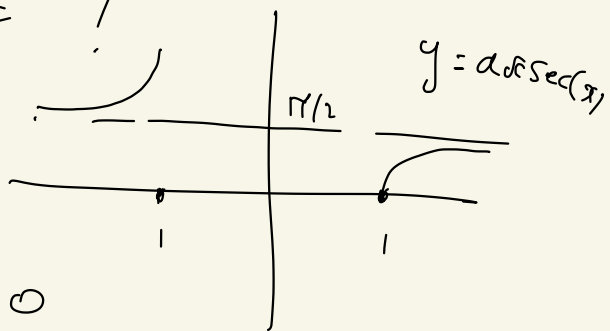
$$\frac{d}{dx} \operatorname{arctan}(x) = \frac{1}{1+x^2}$$

So

$$\int \frac{du}{a^2 + u^2} = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$$

$$\underline{\forall a > 0}$$

$$\bullet \quad \frac{d}{dx} \arcsin = ?$$



$$\frac{d}{dx} \arcsin(x) > 0$$

because $\arcsin$ is increasing.

$$\frac{d}{dx} \arcsin(x) = \frac{1}{(\sin)'(\arcsin(x))}$$

$$\forall x \in (-\infty, -1) \cup (1, +\infty)$$

$$(\sin)' = \left(\frac{1}{\cos}\right)' = -\frac{\cos'}{\cos^2} = \frac{\sin}{\cos^2}$$

[Assume $x > 1$]

$$\sec' = \sec \tan$$

$$\frac{d}{dx} \operatorname{arcsec}(x) = \frac{1}{\sec(\operatorname{arcsec}(x)) \cdot \tan(\operatorname{arcsec}(x))}$$

$$\sec(\operatorname{arcsec}(x)) = x$$

$$\tan(\operatorname{arcsec}(x)) = ?$$

Recall

$$\begin{aligned} \sec^2 &= 1 + \tan^2 \\ &= \tan^2 \end{aligned}$$

$$\tan(y) = \pm \sqrt{\sec(y)^2 - 1}$$

$$= + \quad \text{if } y \in [0, \frac{\pi}{2})$$

because $\tan(y) \geq 0$
if $y \in (0, \frac{\pi}{2})$

Here, we take $y = \operatorname{arcsec}(x)$
 $\in [0, \frac{\pi}{2})$

if $x > 1$

$$\begin{aligned}\text{So } \tan(\operatorname{arcsec}(x)) &= \sqrt{\sec^2(y) - 1} \\ &= \sqrt{x^2 - 1}\end{aligned}$$

Conclusion :

if $x > 1$, we get

$$\frac{d}{dx} \operatorname{arcsec}(x) = \frac{1}{x \sqrt{x^2 - 1}}$$

Similarly, if $x < -1$,

$$\frac{d}{dx} \operatorname{arcsec}(x) = \frac{-1}{x \sqrt{x^2 - 1}}$$

We can combine the two;

$$\frac{d}{dx} \operatorname{arcsec}(x) = \frac{1}{|x| \sqrt{x^2 - 1}}$$

$$\checkmark \quad |x| > 1$$

$$\text{So : } \checkmark \quad a > 0$$

$$\int \frac{du}{u \sqrt{u^2 - a^2}} = \frac{1}{a} \operatorname{arcsec} \left| \frac{u}{a} \right| + C$$

$$\text{Where } u: I \longrightarrow \mathbb{R}$$

$$\text{s.t. } |u| > a$$

$$\underline{\text{Ex : }} \int \frac{1}{x \sqrt{x^8 - 1}} dx = ?$$

$$u = x^4$$

$$du = 4x^3 dx$$

$$\begin{aligned}\frac{du}{u} &= 4 \frac{x^3}{x^4} dx \\ &= 4 \frac{dx}{x}\end{aligned}$$

So

$$\int \frac{1}{x \sqrt{x^8 - 1}} dx = \frac{1}{4} \int \frac{du}{u \sqrt{u^2 - 1}}$$

$$= \frac{1}{4} \operatorname{arcsec}(u) + c$$

$$= \frac{1}{4} \operatorname{arcsec}(x^4) + c$$

—————→

Techniques of integration

To compute $\int f(x) dx$, first try to find an antiderivative of $f(x)$ in a "list" of known antiderivatives.

$$\bullet \int u^n du = \frac{u^{n+1}}{n+1} + C \quad \begin{array}{l} n \in \mathbb{R} \\ n \neq -1 \end{array}$$

$$\bullet \int \frac{du}{u} = \log|u| + C$$

$$\bullet \int \sin(u) du = -\cos(u) + C.$$

$$\bullet \int (\sec(u))^2 du = \tan(u) + C$$

$$\int \sec(u) \cdot \tan(u) du = \sec(u) + C$$

$$\int \tan(u) du = -\log|\cos(u)| + C$$

$$\int e^u du = e^u + C$$

$$\int \frac{du}{\sqrt{a^2 - u^2}} = \arcsin\left(\frac{u}{a}\right) + C$$

$$(a > 0), \quad |u| < a$$

$$\int \frac{du}{a^2 + u^2} = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$$

$$(a > 0)$$

$$\int \frac{du}{u \sqrt{u^2 - a^2}} = \frac{1}{a} \operatorname{arSec} \left| \frac{u}{a} \right| + C$$

Sometimes, we have to make a simple substitution to match one of these integrals

Ex : $\int \frac{x-1}{\sqrt{x^2-2x+3}} dx = ?$

$$u = x^2 - 2x + 3$$

$$dx = (2x - 2)dx = 2(x-1)dx$$

$$\text{So } \int \frac{x-1}{\sqrt{x^2-2x+3}} dx = \frac{1}{2} \int \frac{du}{\sqrt{u}}$$

$$= \sqrt{u} + C$$

$$= \sqrt{x^2-2x+3} + C$$

